



**POLITECNICO
MILANO 1863**

Soft Robotics – Final project

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INTRODUCTION

Project Overview

DESIGN OBJECTIVE

A soft robotic gripper capable of grabbing **different-shaped** objects using **cable-driven** (tendon-like) actuation

PROPOSED SOLUTION

A **3-finger** soft robotic hand capable of secure, **adaptive grasping** in **various environments**, combining **6 key elements**:



Rigid 3D-Printed (FDM)
Palm Frame



Coaxial Double-Spool
Transmission



Palm-Mounted Suction
Cup for Secure Grasp



Elastic Silicone Fingers



2 Antagonistic Tendons
Per Finger (6 Total)



Syringe-Driven Suction
Line

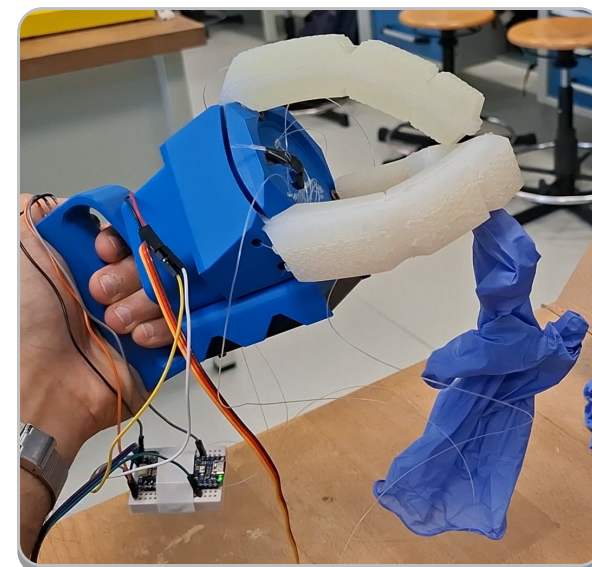
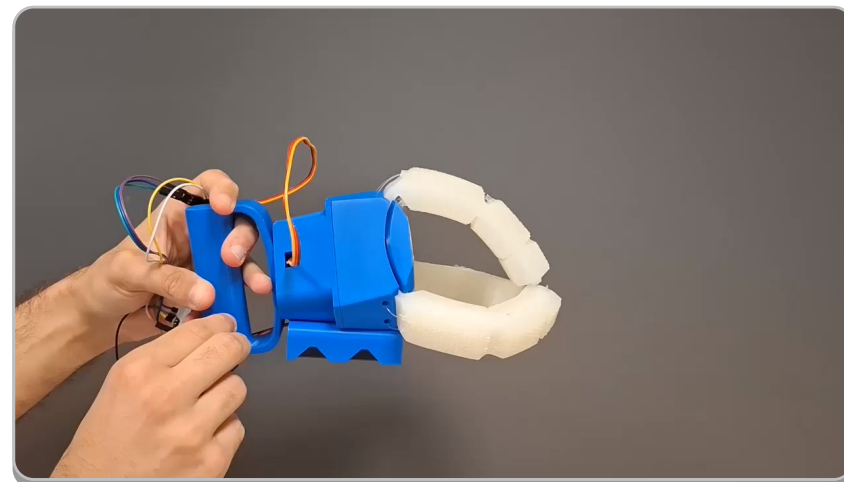
Grabbable Objects

Medium-Sized
Rigid Objects

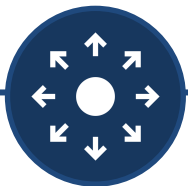
Loose Objects
(Plastic bags,
gloves, etc.)

Dry/Wet Objects

...



Why use Soft Grasping?



Distributed Contact

- Soft fingers **spread contact forces evenly** across the object rather than concentrating them at rigid joints
- Provides the **compliant contact** necessary to safely handle **delicate or irregular** objects
- Utilizes materials with **higher friction** than traditional rigid elements to **improve gripping** and holding capability without increasing force



Large Deformation

- Leverages low elastic modulus materials to achieve significant **bending and stretching** of fingers
- Naturally **wraps around** and passively **adapts** to irregular, unknown, or complex target surfaces
- This **flexibility eliminates** the need for highly accurate object geometry and precise **contact planning**



Simple Actuation and Control

- Achieves highly adaptive grasping with minimal actuation inputs through an **underactuated design**
- **Replaces** heavy, complex multi-degree-of-freedom (DoF) rigid systems
- $\# \text{ motors} < \# \text{ of joints (DoFs)}$
- Drastically **reduces both mechanical and control complexity** while maintaining robust performance

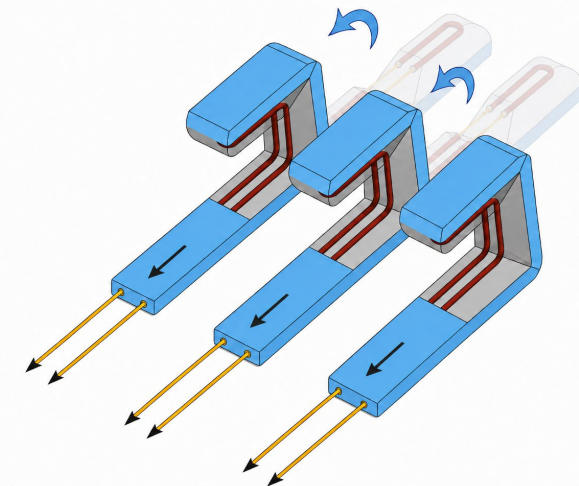
STATE OF THE ART

State of the Art – Tendon-Driven Soft Fingers



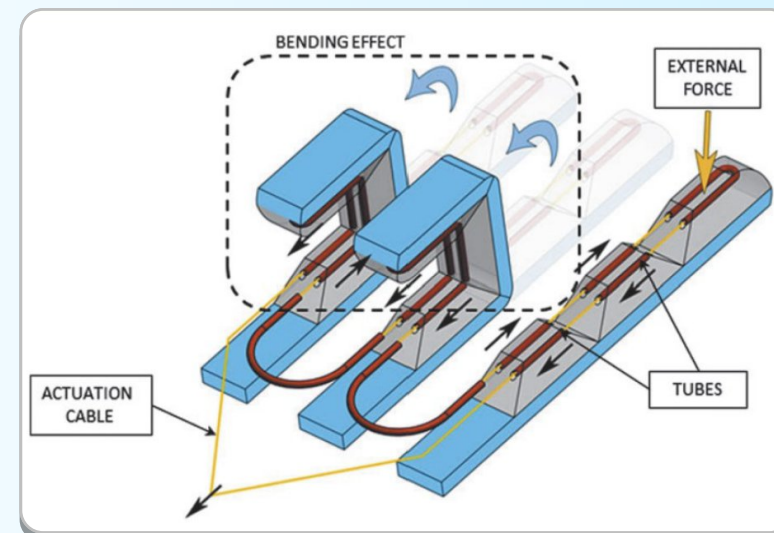
Working Principle

- **Transmits forces** from a motor by shortening or lengthening flexible cables routed directly through the silicone fingers (in series or parallel)
- Produces **natural, adaptive** bending and grasping around **diverse geometries**
- **Eliminates** the need for rigid joint linkages or internal skeletons for each finger

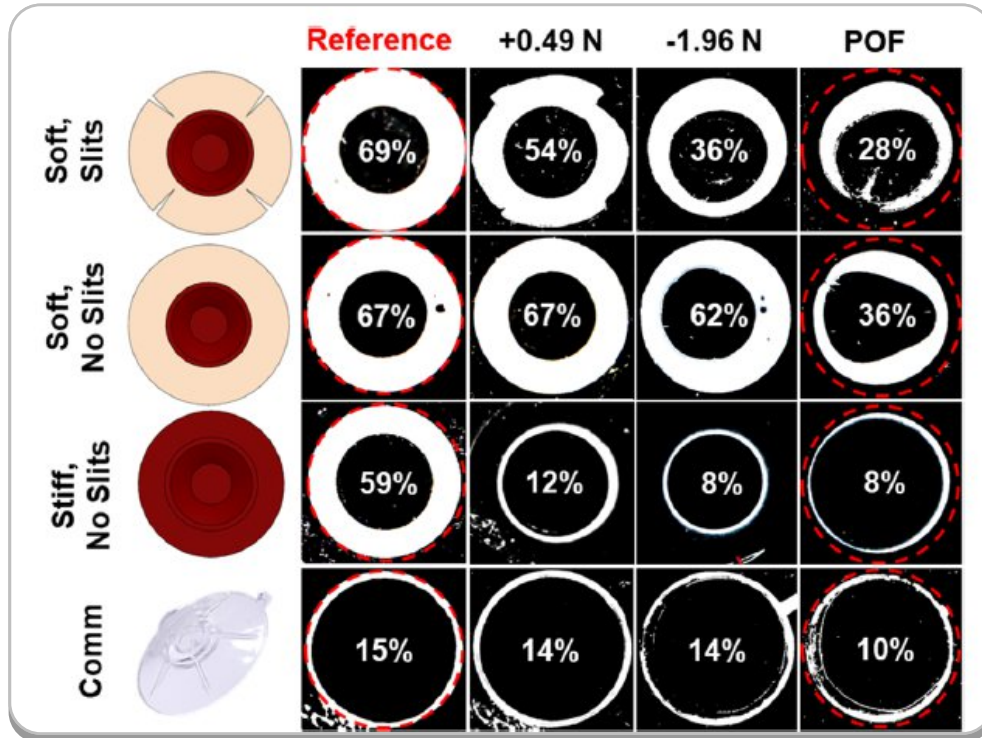


Trade-offs & Limitations

- Hosting cable tensioning motors outside the soft structure provides a **fast** response with **low system inertia, weight, and volume**
- Forces subject the elastomeric fingers to **friction**, mechanical **wear**, and physical **tear** – tendon elasticity wears down -> **hysteresis and creep**
- Highly **nonlinear** (no universal formula)



State of the Art – Suction Cup



["Reversible adhesion to rough surfaces both in and out of water, inspired by the clingfish suction disc", J.A.Sandoval et al., 2019]

WORKING PRINCIPLE

Adhesion is generated through a **pressure differential** between the inner cavity and the surrounding environment

The suction cup is pressed against the object to **expel fluid or air** from the internal chamber and initiate the **attachment seal**

Functions as an **additional adhesion mechanism** that **complements** the gripping forces of the fingers, and can adapt to surface shape and curvature as long as the radial margins/boundaries are firmly sealed

Performance varies depending on the **design parameters**

Suction Cup Geometry

Margin Slits

Material Stiffness

...

SELECTED DESIGN

Soft silicone and **margins without slits** yield the **highest adhesion** for diverse surface textures (incl. rough) in dry environments

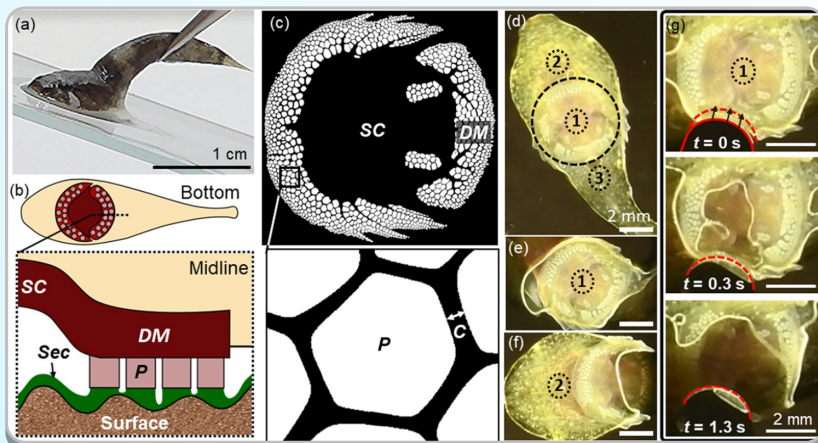
State of the Art – Hexagonal Surface Textures



**Biological
Inspiration**

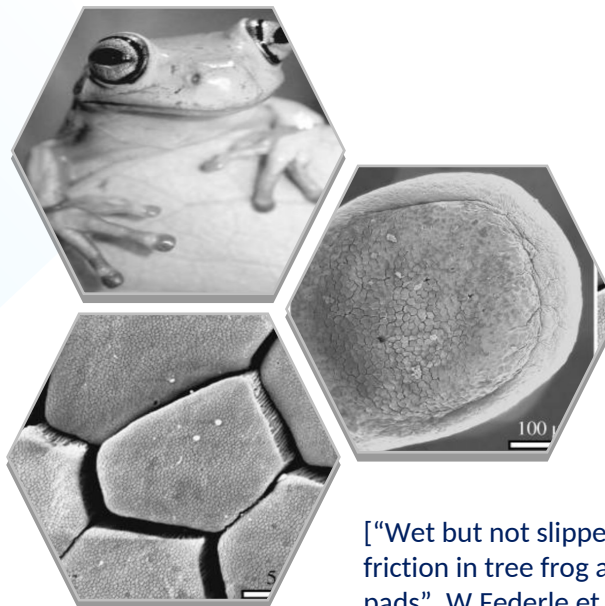
- **Hexagonal microstructures** are found on the attachment pads of the clingfish, tree frog, and bush-cricket
- Channels away boundary fluids from surfaces through micro-grooves between hexagon shapes to **prevent hydrodynamic lubrication (hydroplaning) and the formation of a slippery lubricating fluid film**
- They allow animals to maintain a secure, **high-friction contact interface** even when fully submerged or interacting with highly lubricated wet objects

Clingfish



[“Toward Bioinspired Wet Adhesives: Lessons from Assessing Surface Structures of the Suction Disc of Intertidal Clingfish”, J.A.Sandoval et al., 2020]

Tree Frog



[“Wet but not slippery: boundary friction in tree frog adhesive toe pads”, W.Federle et al., 2006]

Bush Cricket



[“Two functional types of attachment pads on a single foot in the Namibia bush cricket *Acanthoproctus diadematus*”, C. Grohmann et al., 2015]

State of the Art – Combination of Soft Gripper and Suction

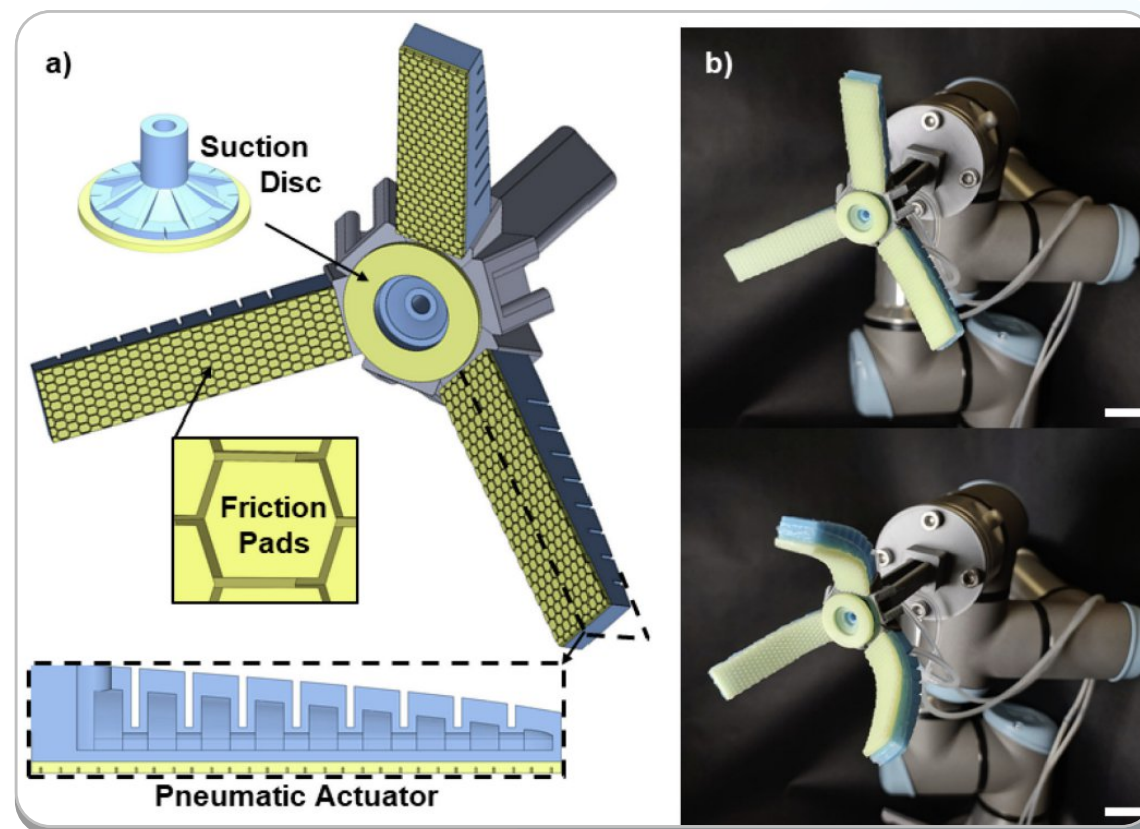
CHALLENGING WET ENVIRONMENTS

Boundary fluids act as a **lubricant** between soft fingers and target surfaces, reducing friction and causing hydrodynamic slipping during object manipulation

HYBRID APPROACH

- **Combination** of texture micro-channels and central palm suction cup to divert boundary fluids and re-establish direct between fingers and surface material
- **Two complementary force mechanisms** where the suction cup handles normal pull-off resistance and the fingers provide lateral shear stability

Approach achieves **higher tolerance** under **combined** normal and shear loads *than either suction or soft fingers can provide alone*



[“Combining Suction and Friction to Stabilize a Soft Gripper to Shear and Normal Forces, for Manipulation of Soft Objects in Wet Environments”, J.A.Sandoval et al., 2022]

DESIGN

From reference to our idea

STARTING POINT

Underactuated soft gripper: tendon-driven soft finger, one motor, one DOF, passive elastic recovery.

Bio-inspired design solutions: suction disk (like clingfish) with surface texture for enhanced grip + shape of human finger



OUR IDEA IMPLEMENTATION

Three-finger gripper with two antagonistic tendons per finger → active bidirectional bending (via Double-spool mechanism).

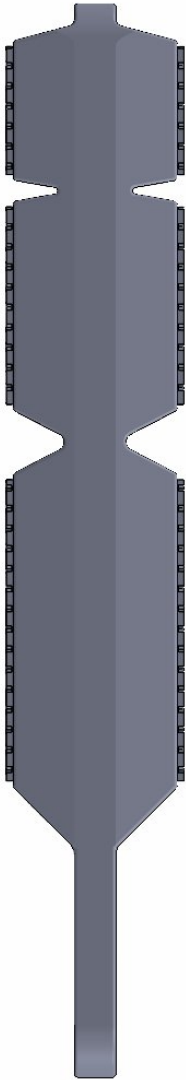
Palm suction cup for enhanced grasping reliability.

Hexagonal texture on finger surfaces for improved wet grip.

Design - Finger



FRONT



EDGE



Material & Geometry

- 3 soft silicone fingers in a 120° radial configuration
- Symmetric shape allowing bidirectional bending

Distribute contact forces evenly across three points



Surface Texture

- Embedded a biomimetic array of hexagonal microstructures onto the primary contact surfaces

Prevent hydrodynamic slipping in wet conditions



Three Finger Segments

- Segmented with decreasing height from base to tip
- Tapered profile along the length of the finger

Promotes natural curling and improves object wrapping



Dimensions

- Overall finger length 80 mm with 20 mm base width and 24 mm finger width
- Phalange lengths of 45.24 mm, 30.35 mm, 19.30 mm

Scaled up from natural human proportions for target objects



Geometric Angles

- Set inter-phalangeal angles at 20° and 55° to balance bending range of motion and contact area
- 7.5° sidewall inclination angle

Prevents overlap of fingertips during full closure for grasping

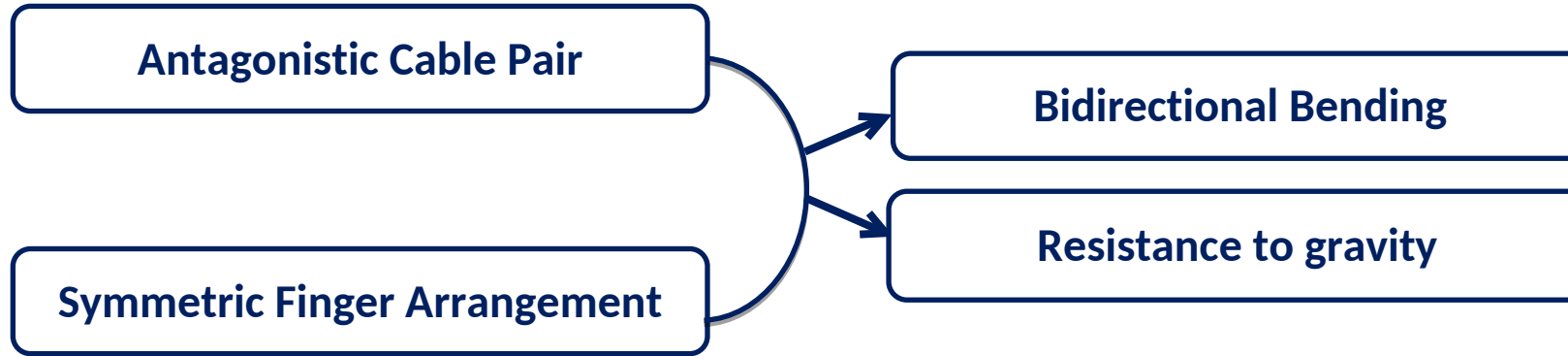


Frame Integration

- Directly embedded the silicone finger bases into the rigid 3D-printed FDM frame

Provides an anchor point with low compliance at the base

Design – Antagonistic Cable Pair



BIOLOGICAL INSPIRATION: Muscular hydrostats

Adaptable bending inspired by **antagonistic muscles**; artificially reproduced with antagonistic tendon-based actuation.



Squid and octopus arms



Elephant trunk



Vertebrate tongue



Design – Actuation

Coaxial double-spool design

Two coaxial spools with **oppositely wound** cable pairs, as one set of cables winds, the opposing set unwinds.

Parallel wire arrangement

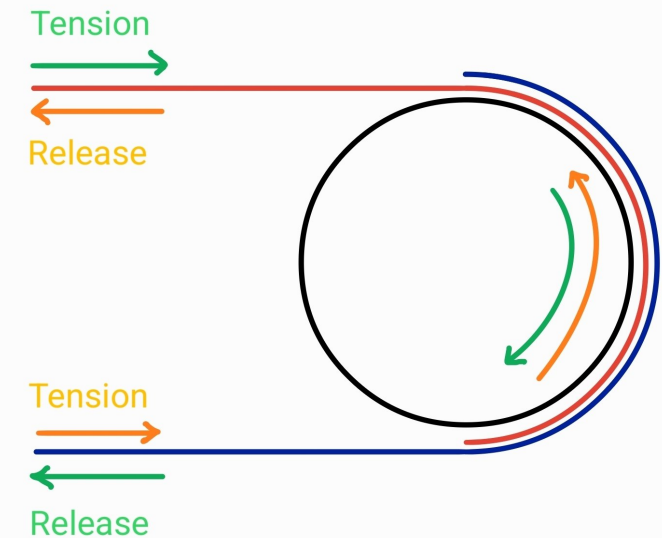
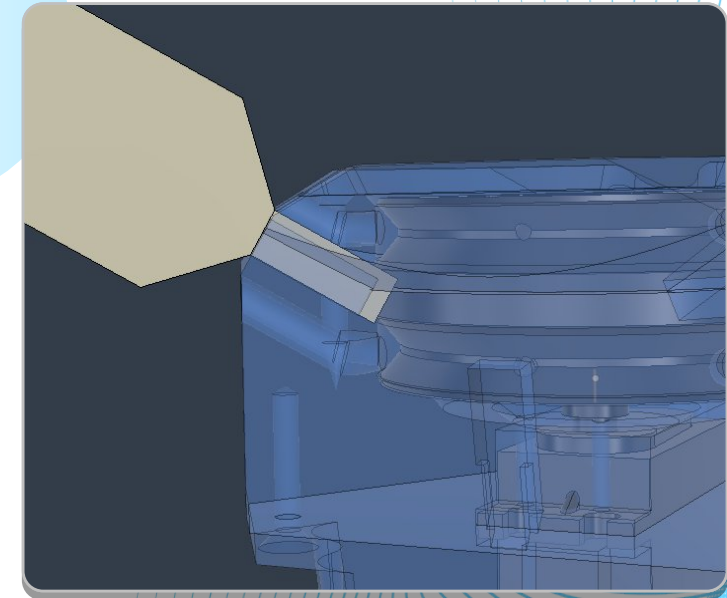
To **mitigate effects of built-up friction** along the whole wire line a series arrangement would have.

Single DOF control (1 rotational input)

One spool rotation angle drives 6 cables and 3 fingers simultaneously, achieving full open/close behavior with minimal actuation and control hardware.

Arduino-driven actuation

Servo motor is driven through a potentiometer.



Design – Palm Suction Cup

Moldable Design

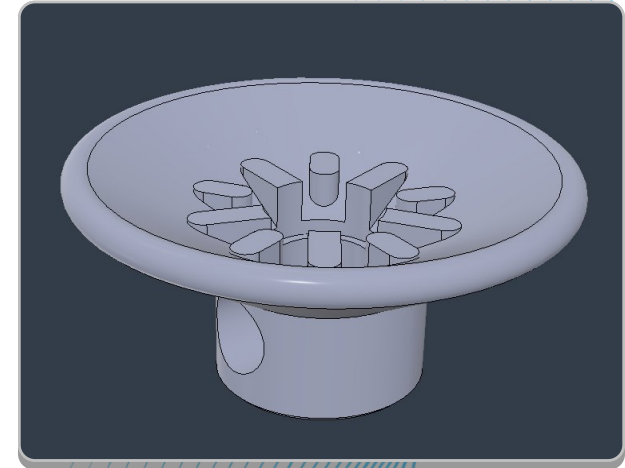
The suction cup was designed considering the molding process and flow of silicone

Ribbed Center

The ribbed center achieves **structural integrity, controlled deformation, controlled airflow**

Pressure Sensing

The suction line is integrated with a pressure sensor.



Design – Frame

Compact Design

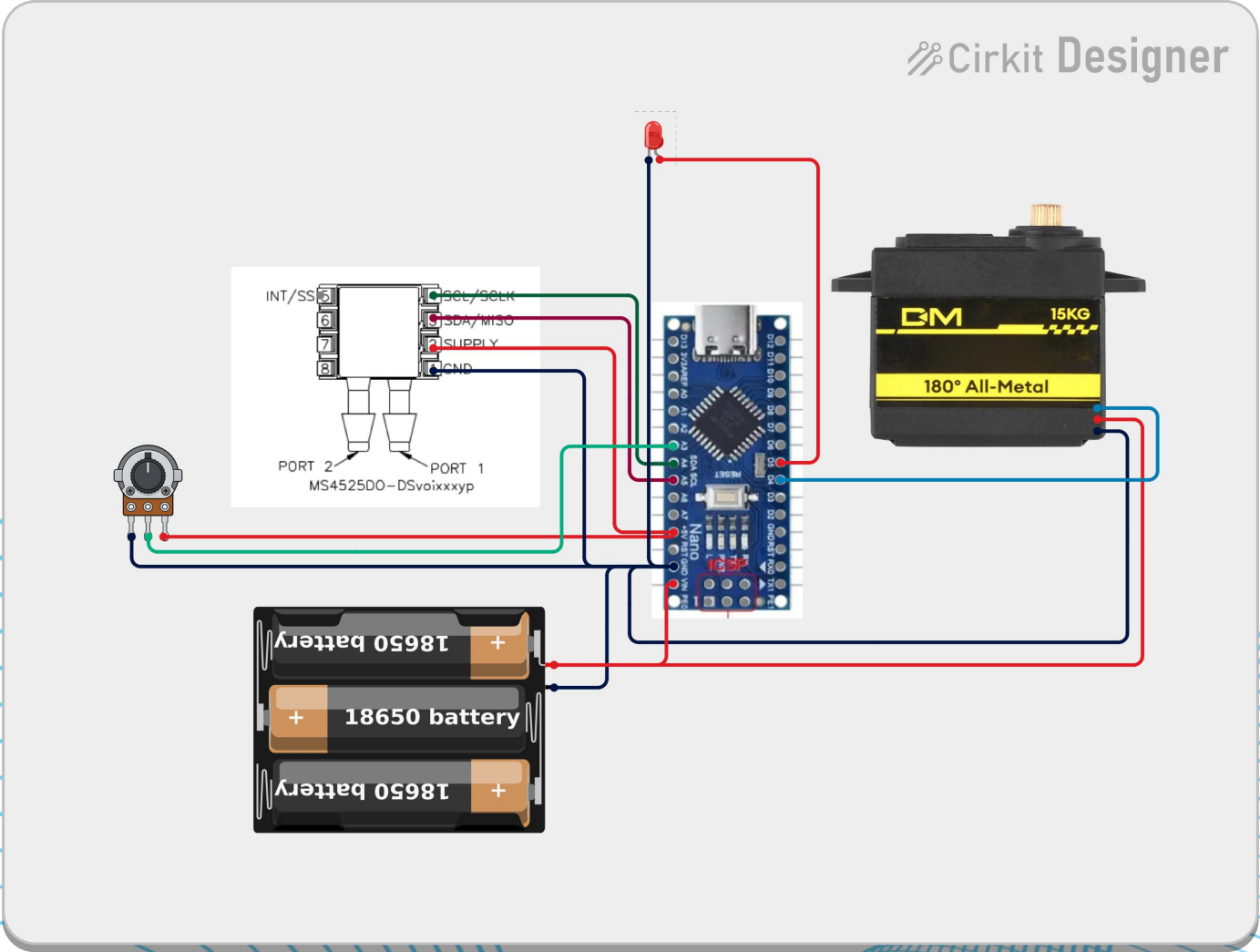
The frame geometry is designed to be as compact as possible to increase reachability of the gripper

Modular Design

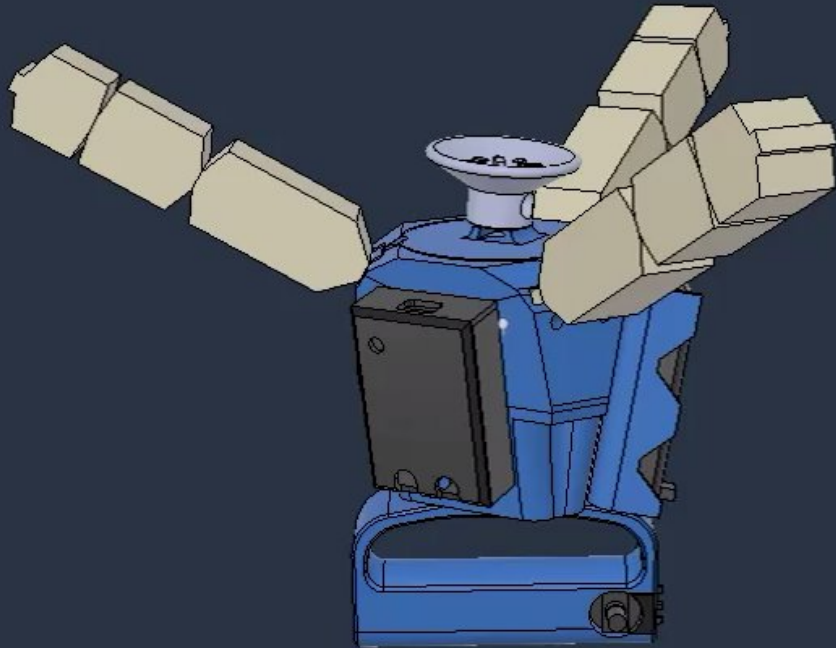
The frame is modular and allows for future customization. The frame can be attached with a handle or can be attached to a robot arm. The palm of the gripper houses a slot for an optional suction cup. There are also slots for the battery and electronic components along the sides of the frame.



Hardware implementation



Robot Architecture



1 Rigid 3D-printed frame

2 Silicone elastomer fingers (×3)

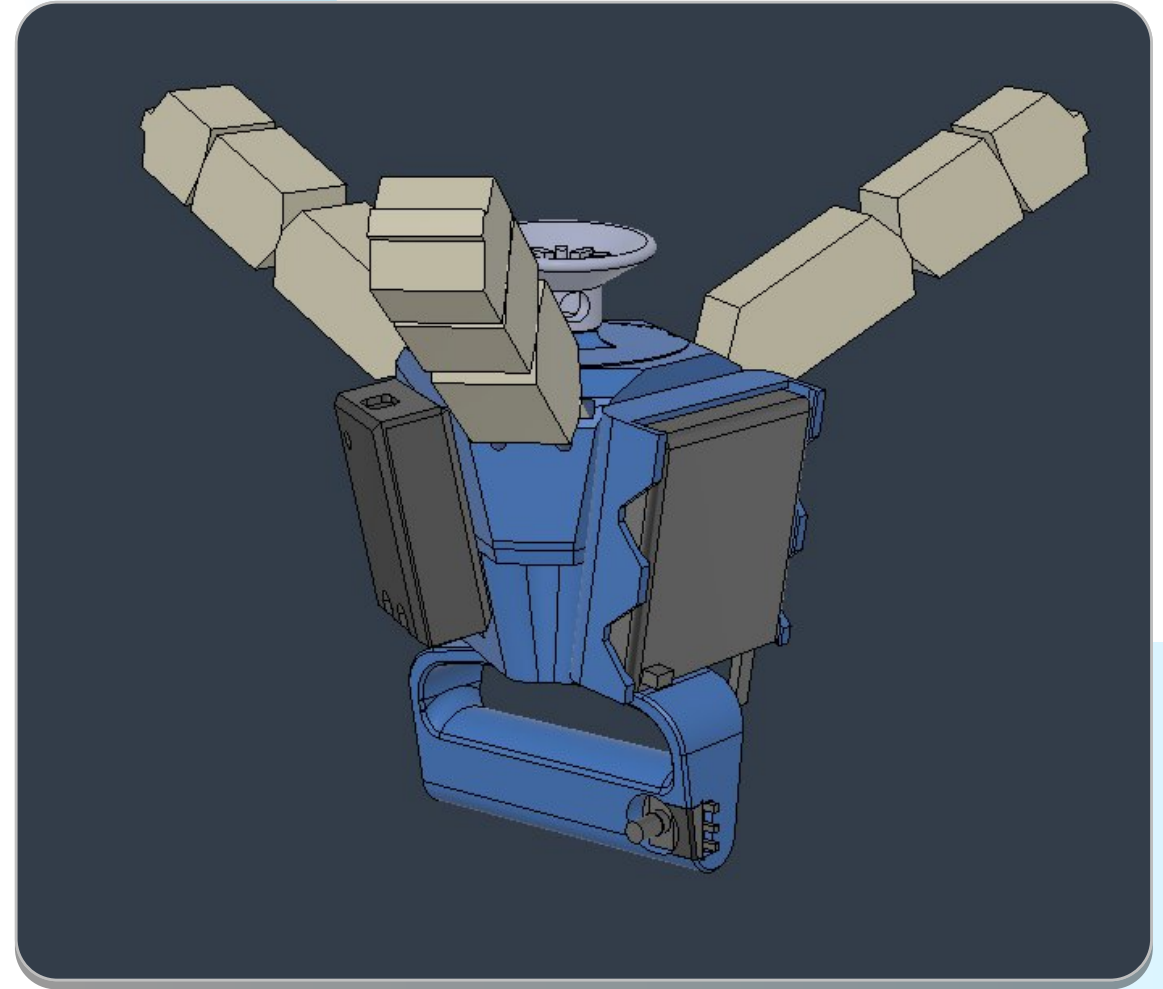
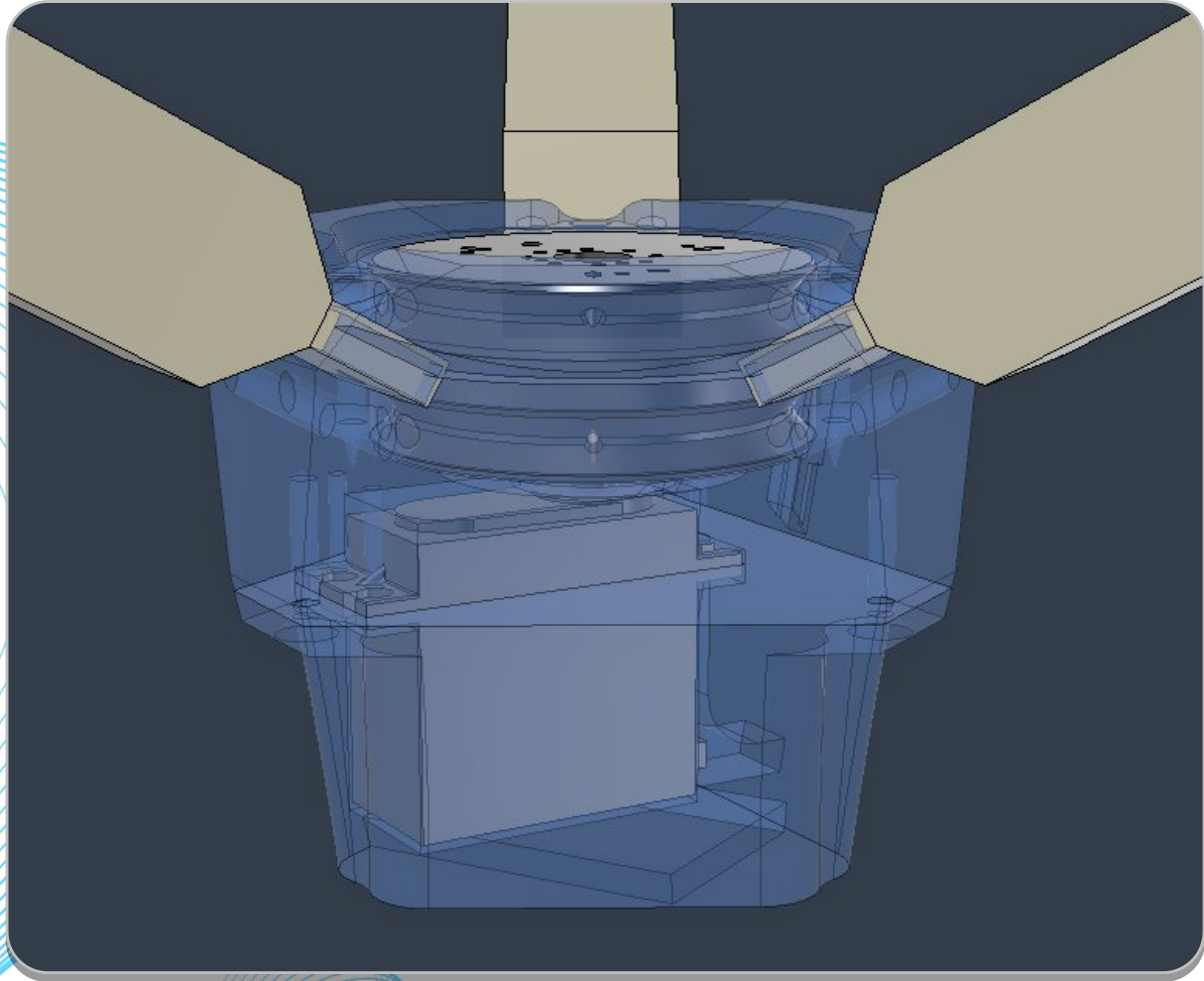
3 Antagonistic tendons
(×2 per finger)

4 Coaxial double spool

5 Palm suction cup with syringe-
driven vacuum line

6 Pressure sensor

Design - Frame



MANUFACTURING

Mold-Based Silicone Casting: Finger

THE RESULT



1. Mold CAD design and 3D printing

3. Weighing and mixing of two-component silicones

5. Partial curing in separated molds

7. Demolding and flash removal

2. Insertion of cable guides and mold release spray application

4. Individual casting: pouring in the molds

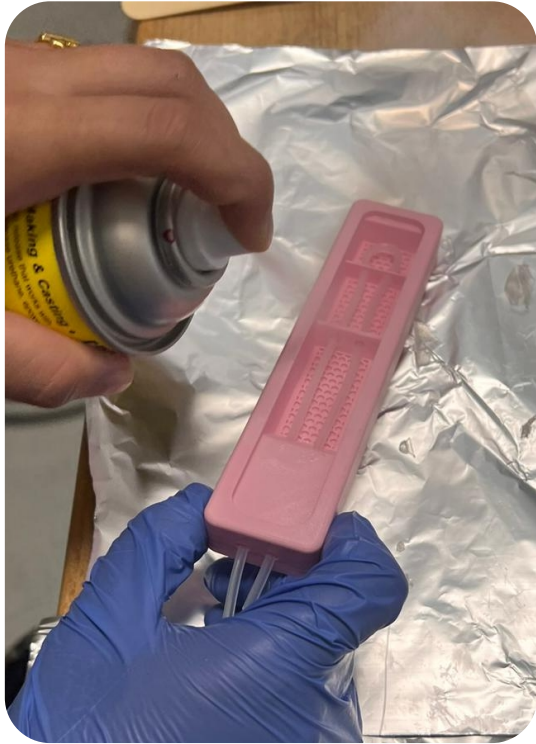
6. Molds overlapping

8. Tendon insertion



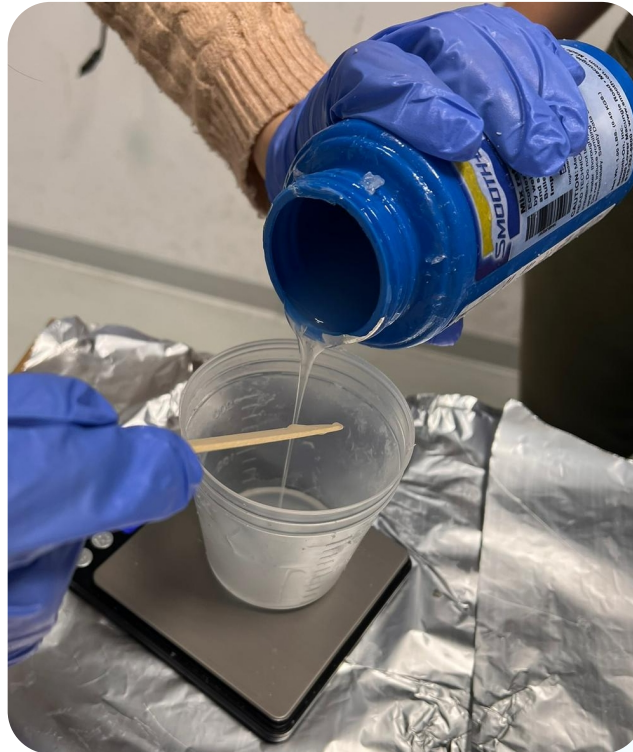
Mold-Based Silicone Casting: Finger In-Lab Fabrication

2.



- Insertion of cable guides to reduce cable friction and constrain routing path
- Application of mold release spray

3.



Mixing two-component silicone: before Ecoflex, then Dragon Skin

4.



Casting the two molds separately: Ecoflex followed by Dragon Skin

6.



Molds overlapping and final curing

Mold-Based Silicone Casting: Finger In-Lab Fabrication



First prototype of a half of a finger

LAYERING OF TWO SILICONE TYPES

Ecoflex 00-35 FAST

- Lower Young's modulus
- ☾ softer silicone with lower structural support
- à easier deformation



Entire finger demolded

Dragon Skin 10

- Higher Young's modulus
- ☾ stronger and less compliant silicone
- ☾ provides body and shape stability

Silicone dosage per mold	
Ecoflex 00-35 FAST	Dragon Skin 10
25 g	12 g

Mold-Based Silicone Casting: Suction Cup

5-STEP PROCESS:

- Modular mold: CAD design and 3D printing
- Mixing of two-component Ecoflex silicone
- Casting
- Curing
- Demolding and flash removal

THE RESULT:

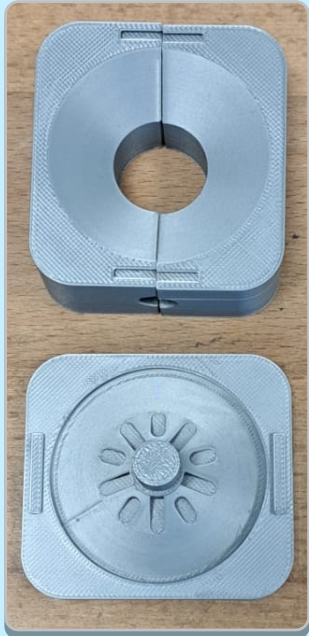


Silicone dosage

Ecoflex 00-35 FAST	12 g
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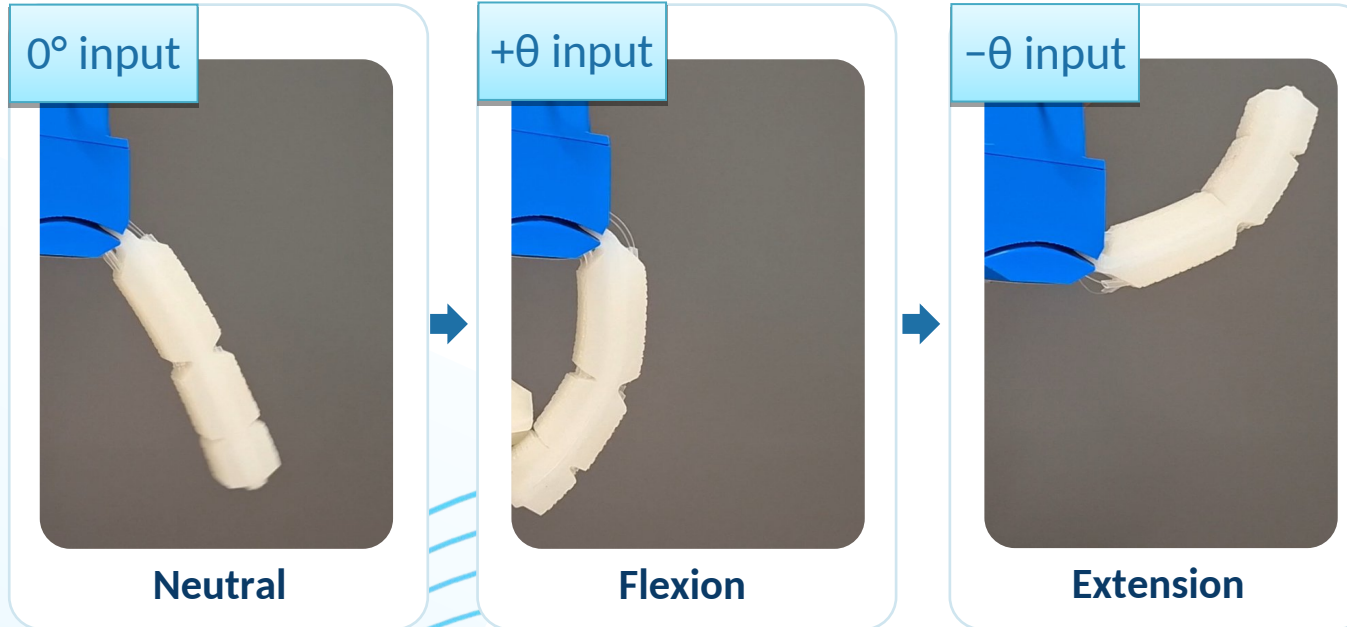
3D printed mold



TESTING & VALIDATION

Single Finger Motion Test

AIM: Evaluating active bidirectional motion of a single antagonistic tendon-driven finger.



Repeated flexion-extension cycles under identical motor commands

Evaluation criteria

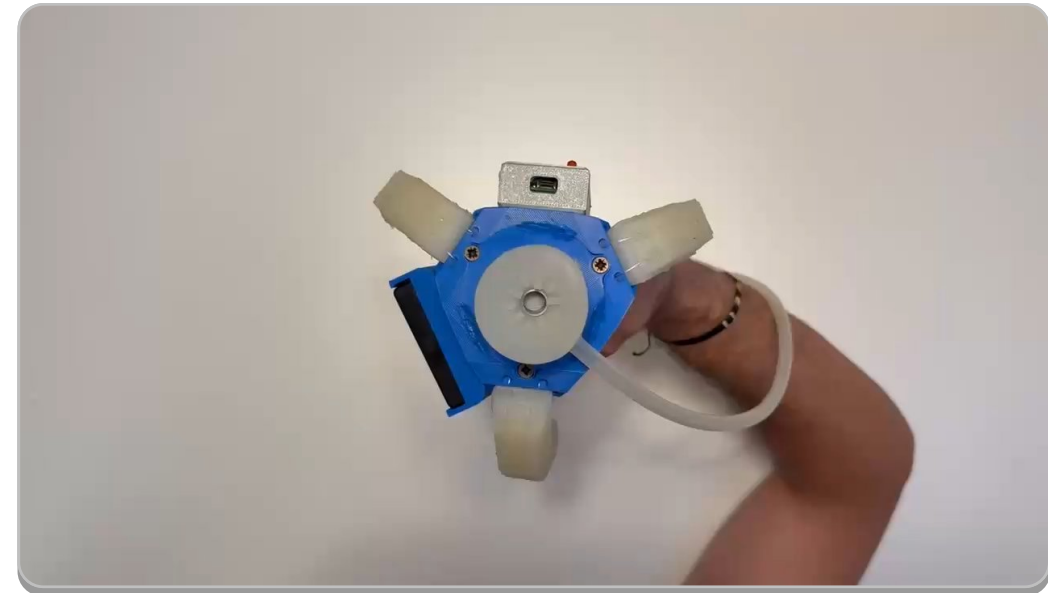
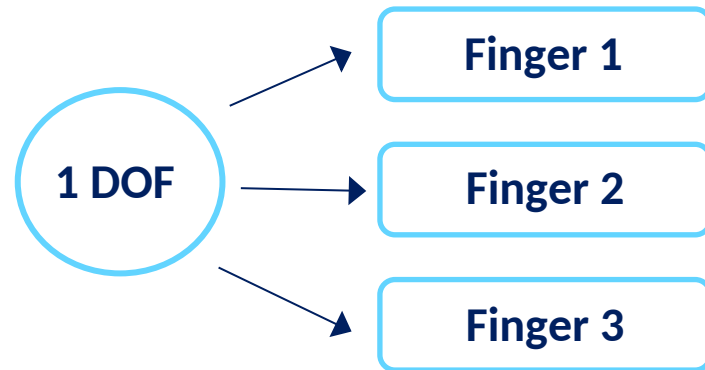
- ① **Bidirectional bending**
Active flexion & extension
- ② **Repeatability**
Consistent motion across repeated cycles
- ③ **Return behavior**
Returns close to neutral after unloading

The antagonistic tendon mechanism enables reliable active bidirectional bending of a single soft finger.

Three-Finger Coordination Test

AIM: Validation of coordinated three-finger actuation using a single rotational input.

MECHANICAL COUPLING: Coordinated Three-Finger Motion



EVALUATION CRITERIA

Motion onset

All fingers begin moving simultaneously.

Direction

Consistent opening and closing direction.

Final curvature

Comparable finger curvature after actuation.

Delay/asymmetry

Minor differences caused by friction and tendon pretension.

✓ **The double-spool mechanism successfully synchronizes three-finger motion using a single rotational DOF**

Adaptive Grasping Test

AIM: Evaluating external, internal and adaptive grasping with compliant bidirectional fingers.

EXTERNAL ENVELOPING GRASP



INTERNAL EXPANDING GRASP



SHAPE-ADAPTATION CHALLENGES

DEFORMABLE / FLEXIBLE OBJECT



IRREGULAR OBJECT



MISALIGNED APPROACH



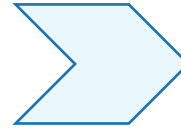
- ✓ Bidirectional actuation enables both inward enveloping and outward expanding grasps, while compliance supports deformable, irregular and misaligned objects.

Suction-Assisted Grasping

AIM: Evaluating palm suction as an auxiliary adhesion mechanism for stable object holding.



BASELINE



ENHANCED

COMPLEMENTARY ROLES & SEAL-DEPENDENT PERFORMANCE

Soft fingers

Distributed contact
+ shear stability

Palm suction

Normal adhesion

Evaluate

- seal quality
- with / without suction
- pull-off resistance
- leakage or misalignment

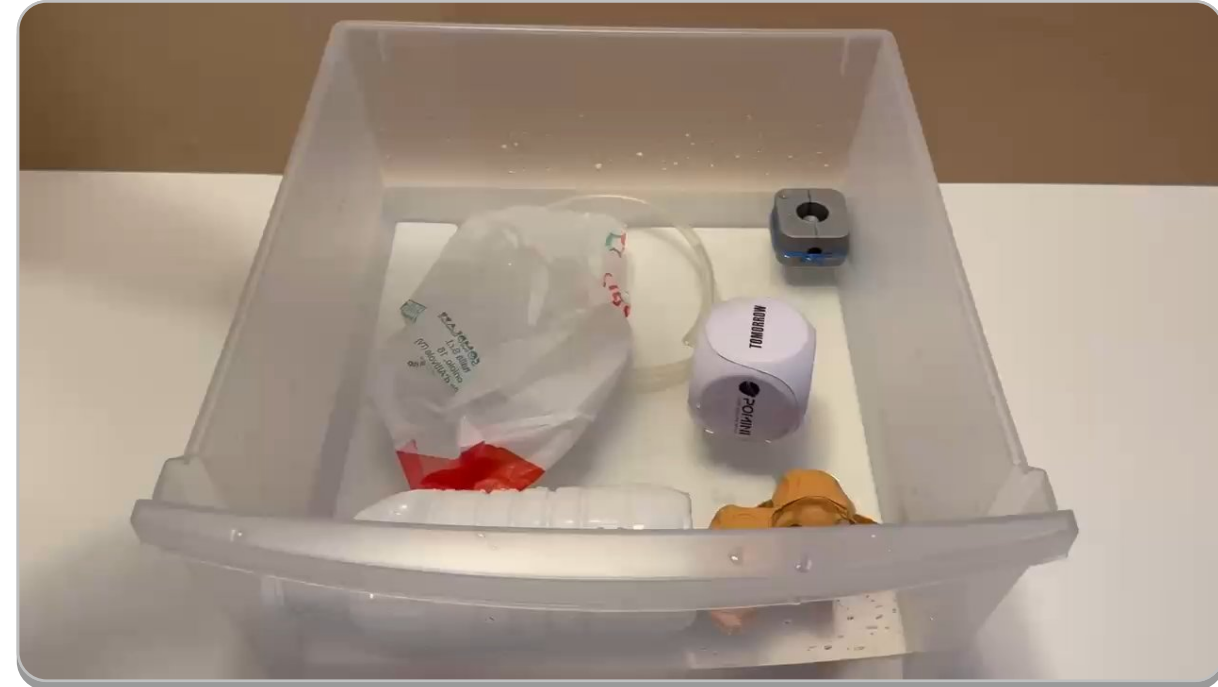


Wet Environment Performance

AIM: Validating performance in wet conditions, the target scenario that motivated the hexagonal surface texture.



Single object retrieval — reduced grip, increased weight from water



Multi-object retrieval — plastic bags, waste, mixed debris

✓ Hexagonal texture prevents hydroplaning, enabling reliable grip on wet and submerged objects.

Validation Against Design Objectives

AIM: Connecting the experimental observations to the original prototype requirements.

DESIGN OBJECTIVE	VALIDATION OUTCOME
01 Active bidirectional bending	ACHIEVED
02 One-input three-finger coordination	ACHIEVED
03 Adaptive grasping across geometries	ACHIEVED
04 Suction-assisted holding	ACHIEVED

The prototype demonstrates the proposed low-complexity hybrid grasping concept: compliant fingers provide adaptation, the double spool provides coordination, and suction adds holding support when sealing is possible

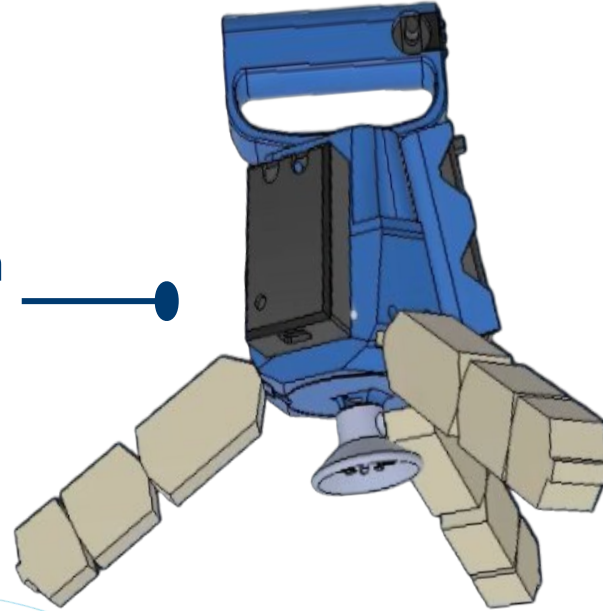
CONCLUSION & FUTURE WORK

Conclusion

RESULT: Low-complexity soft grasping through compliance, bidirectional actuation and suction.

Antagonistic actuation

Active bidirectional flexion and extension.



Underactuation & mechanical coupling

One rotational DOF coordinates six tendons and three fingers.

Compliance & bioinspired adhesion

Adaptive external, internal and suction-assisted grasping.

COMPLETE SOFT ROBOTIC GRIPPER

The prototype demonstrates how compliance, tendon-driven underactuation and mechanical coupling can expand grasping capability while keeping actuation and control complexity low.

Limitations & Future Work

① Transmission variability

Cable friction, slack and pretension imbalance can introduce delay and small inter-finger differences.



① Refined tendon routing

Use low-friction guides, adjustable tensioners and improved anchoring to balance the cable pairs.

② Fabrication tolerances

Casting bubbles, channel inaccuracies and silicone variability affect stiffness and bending symmetry.



② More repeatable soft fabrication

Improve mold accuracy, vacuum-degas silicone and standardize curing/demolding conditions.

③ Seal-dependent suction

Suction performance depends on surface geometry and is currently driven by a manual vacuum source.



③ Autonomous suction control

Integrate a compact pump with pressure sensing to detect sealing and regulate vacuum automatically.

④ Qualitative validation

Performance is mainly evaluated by observing videos and grasp success, without force benchmarks.



④ Quantitative closed-loop testing

Measure bending angle, payload, pull-off force, success rate and durability over repeated cycles.

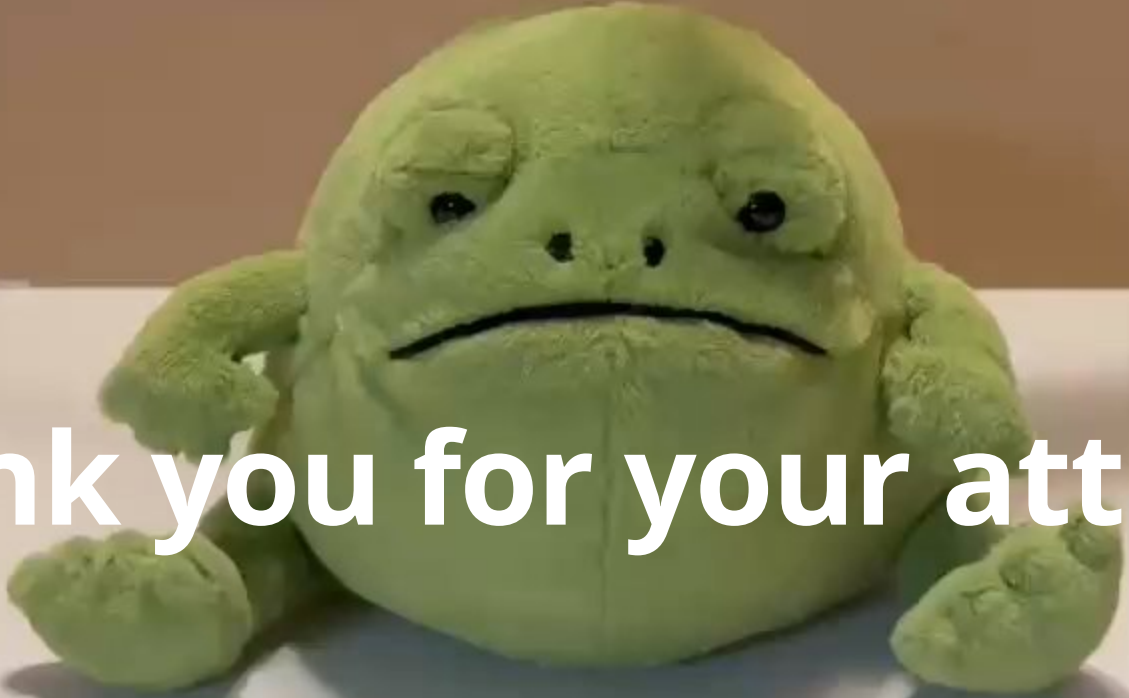
Next Steps and Future Works



Reduce mechanical variability and automate suction



Waterproofing and validation in wet environments

A green plush frog with a sad expression, sitting on a white surface against a brown background. The frog has large, dark eyes and a downturned mouth. The text "Thank you for your attention!" is overlaid in white.

Thank you for your attention!